

# Network Science

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## Lecture 07: Small-World Networks

5942UE Network Science

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# Small-World Networks

This lecture asks why short paths are common and when decentralized navigation is possible.

## Focus

- the small-world phenomenon
- Watts–Strogatz structure and randomness
- decentralized search
- empirical validation and the Web as a directed information network

# Module 1

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## Small-World Phenomenon

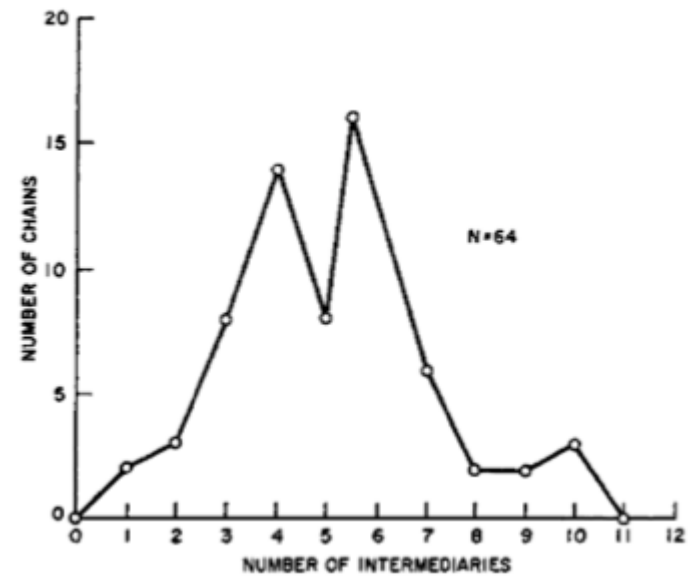
# Guiding Question

Why are distances in large networks often surprisingly short?

# Formal Intuition

The small-world claim is that typical shortest-path distances grow **slowly** relative to network size.

# Milgram



# Core Result

$$d_{\text{typical}} \propto \log |V|$$

# Interpretation

The small-world claim is **not** that everyone is directly connected. It is that typical distances grow slowly even in very large networks.



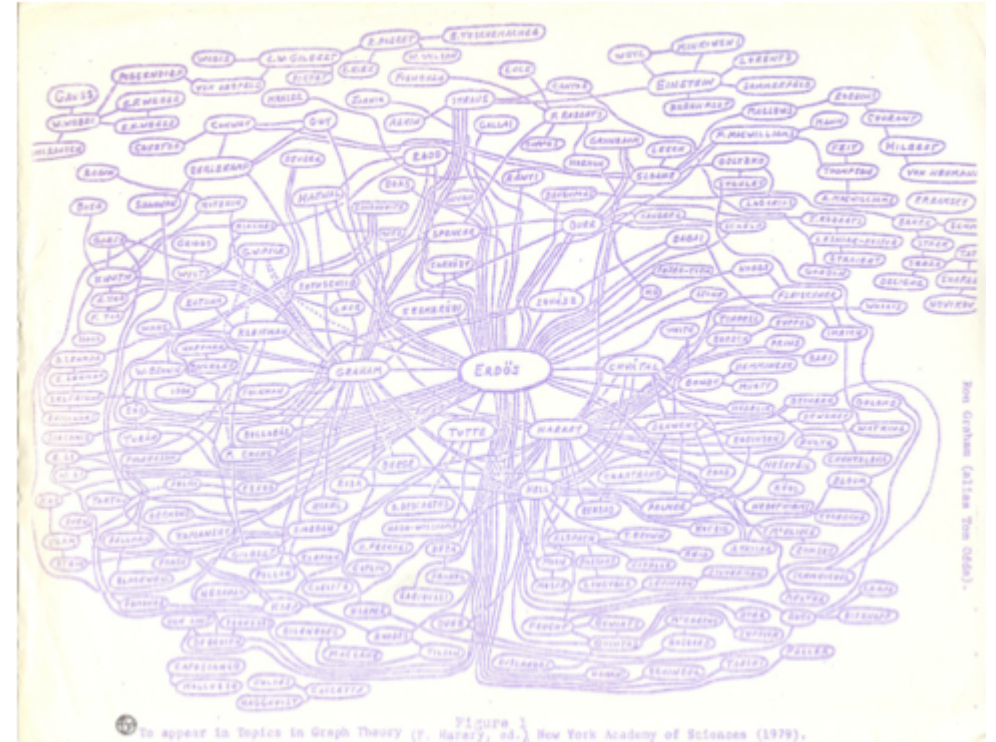
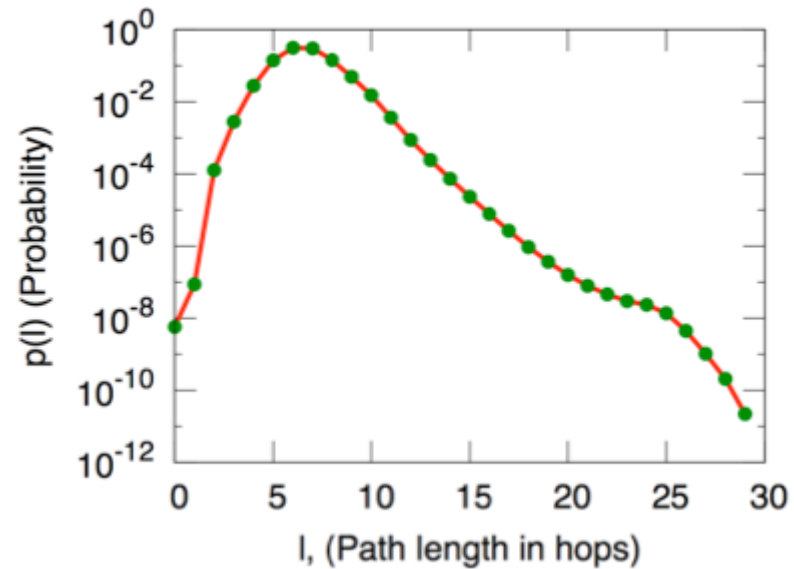


Figure 2.12: Ron Graham's hand-drawn picture of a part of the mathematics collaboration graph, centered on Paul Erdős [189]. (Image from <http://www.oakland.edu/enp/cgraph.jpg>)

Large networks can still be navigably small in terms of path length.

# Module 2

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## Structure and Randomness

# Guiding Question

How can a network be both highly clustered and still have short paths?

# Formal Intuition

Clustering and short average path length are **not mutually exclusive**.

The key is that a few **long-range links** can drastically reduce distances.

# Formation of Short Paths

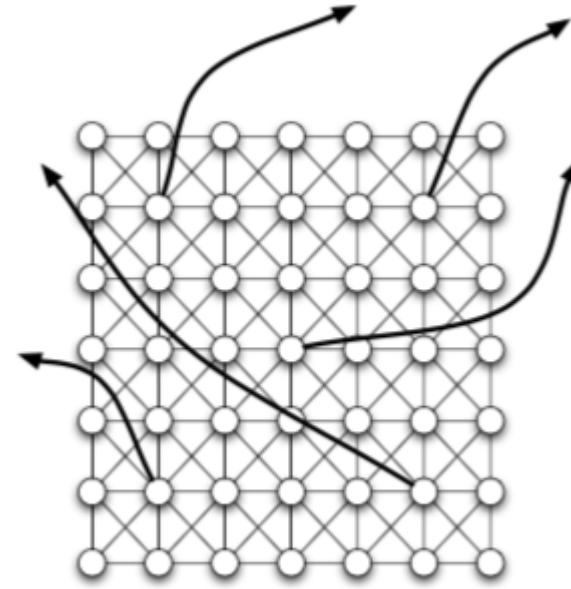
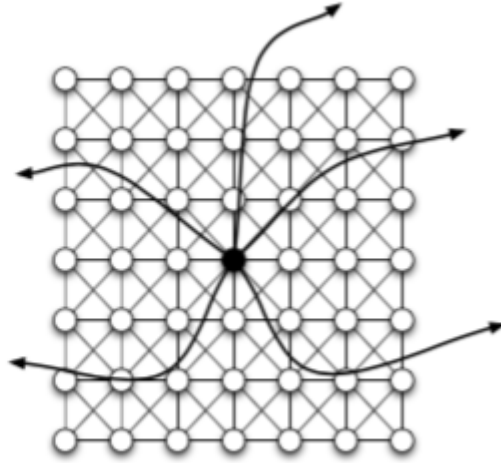
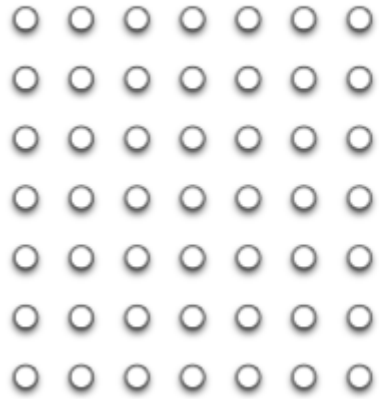


(a) *Pure exponential growth produces a small world*



(b) *Triadic closure reduces the growth rate*

# Watts–Strogatz



# Interpretation

Local links preserve **clustering**. A small amount of random long-range linkage collapses **path length** dramatically.

A tiny dose of randomness can make the world small without destroying local structure.

# Module 3

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## Decentralized Search



# Guiding Question

Why do short paths not automatically imply that decentralized actors can find them?

# Problem

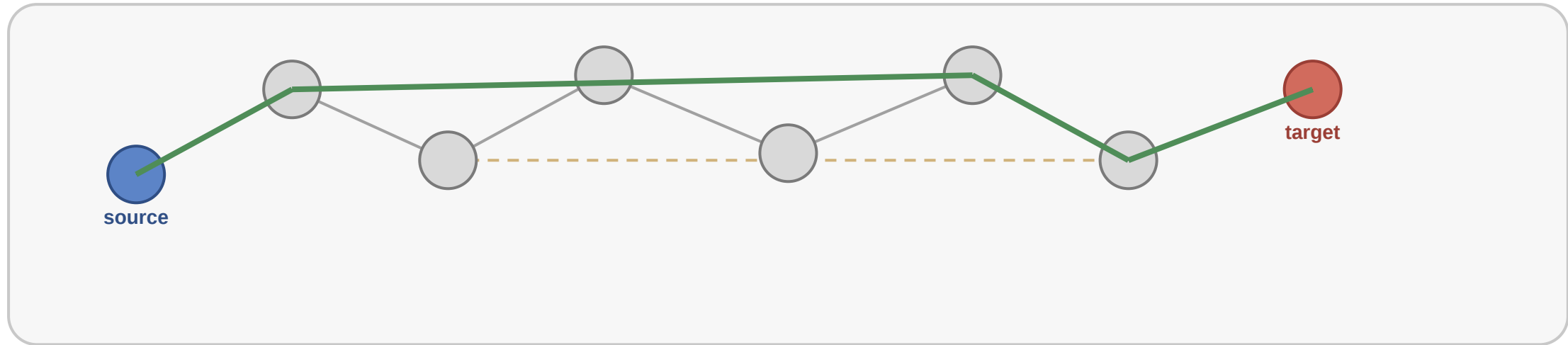
Milgram-like routing is **not** breadth-first search. Each actor sees only local information.

# Formal Intuition

Decentralized search succeeds when the network places long-range links in a way that **local decisions can exploit**.

# Kleinberg-Style Idea

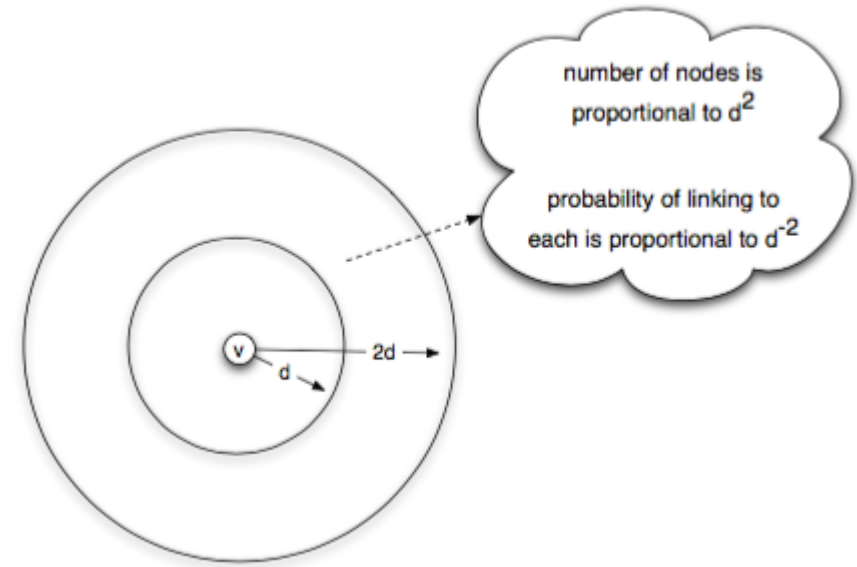
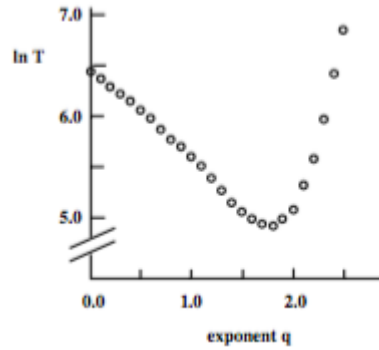
Decentralized search uses local views, not global shortest paths



Green path is found by local forwarding; dashed links are long-range shortcuts that make local search possible.

Long-range links should not be uniformly random. They should be distributed **across scales**.

# Evidence



Efficient search requires not only short paths, but the right geometry of long-range ties.

# Module 4

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## Empirical Analysis

# Guiding Question

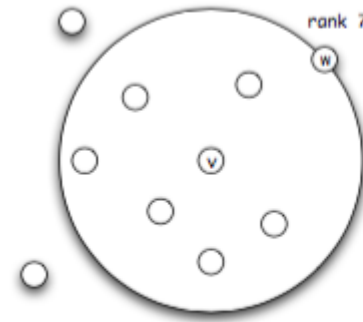
Does the decentralized-search model show up in real data?

# Formal Intuition

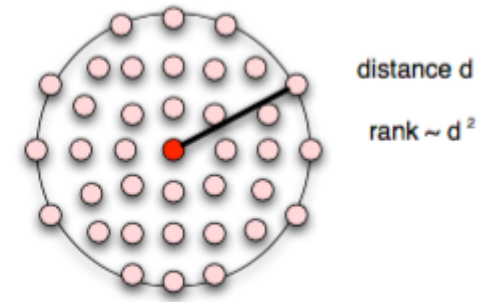
The empirical question is whether real networks organize long-range links across scales in the way search models predict.



# Geographic Rank

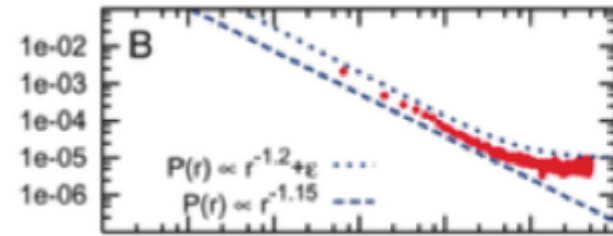


(a)  $w$  is the 7<sup>th</sup> closest node to  $v$ .

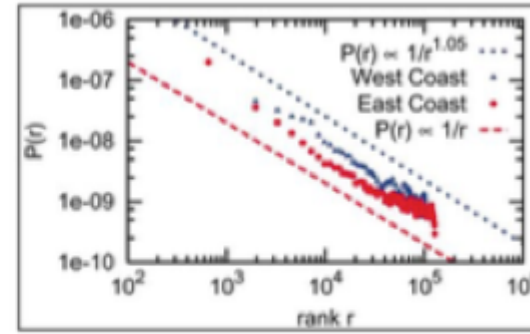


(b) Rank-based friendship with uniform population density.

# Evidence

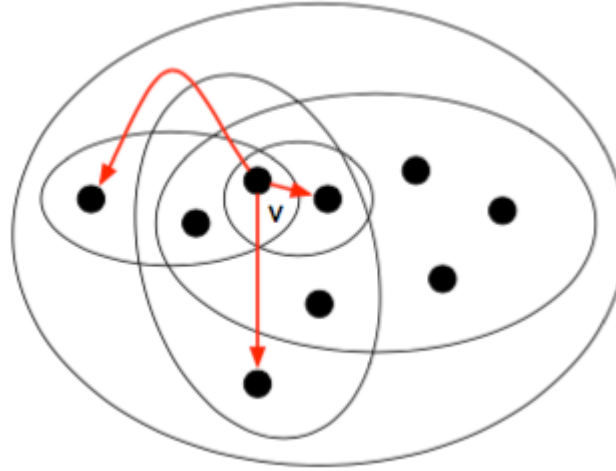


(a) Rank-based friendship on LiveJournal



(b) Rank-based friendship: East and West coasts

# Beyond Geography



Empirical networks often organize weak ties in ways that support decentralized navigation, even when distance is social rather than geographic.

# Module 5

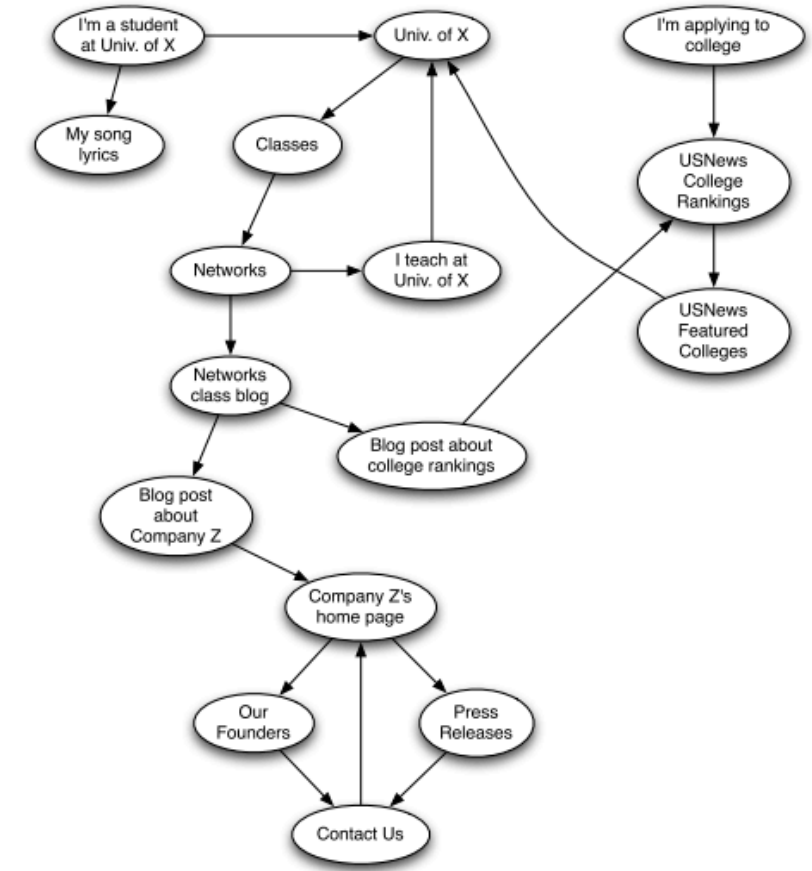
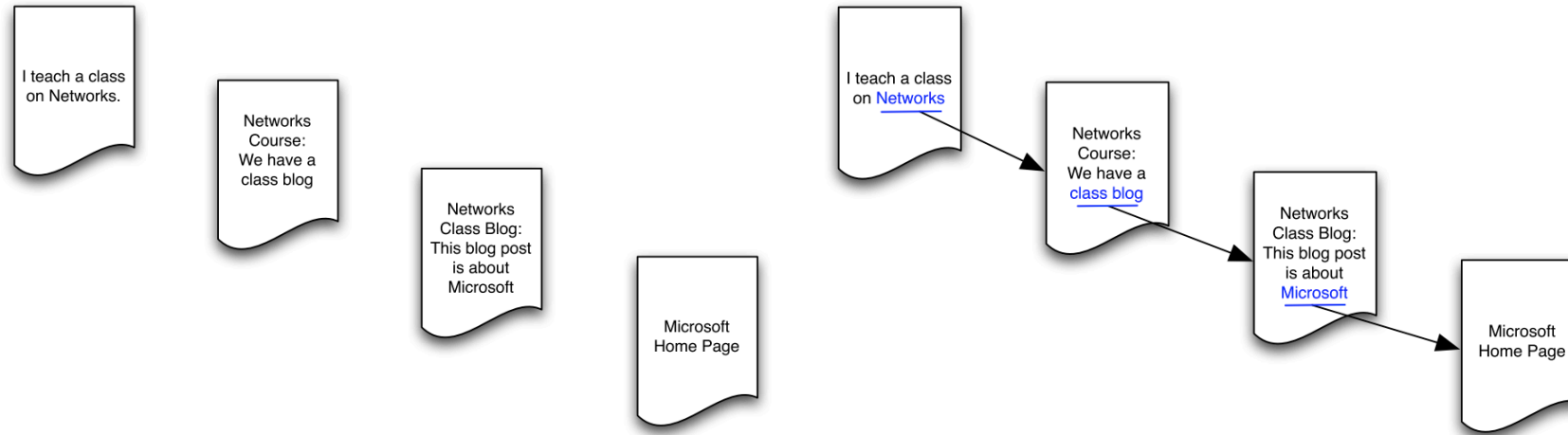
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## The Web as a Directed Network

# Guiding Question

What changes when we move from social networks to the Web?

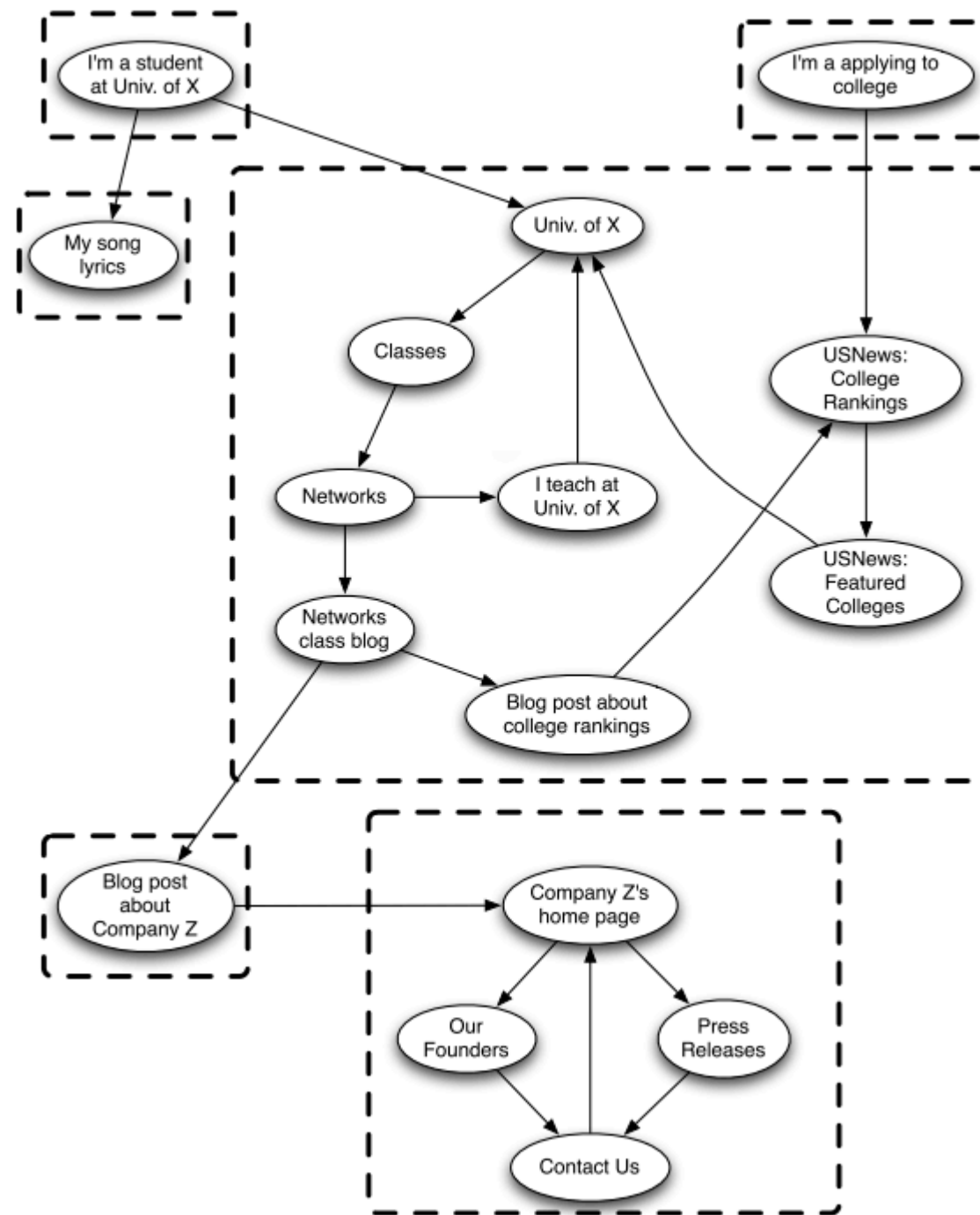
# From Hypertext to Directed Graphs



# Strong Connectivity

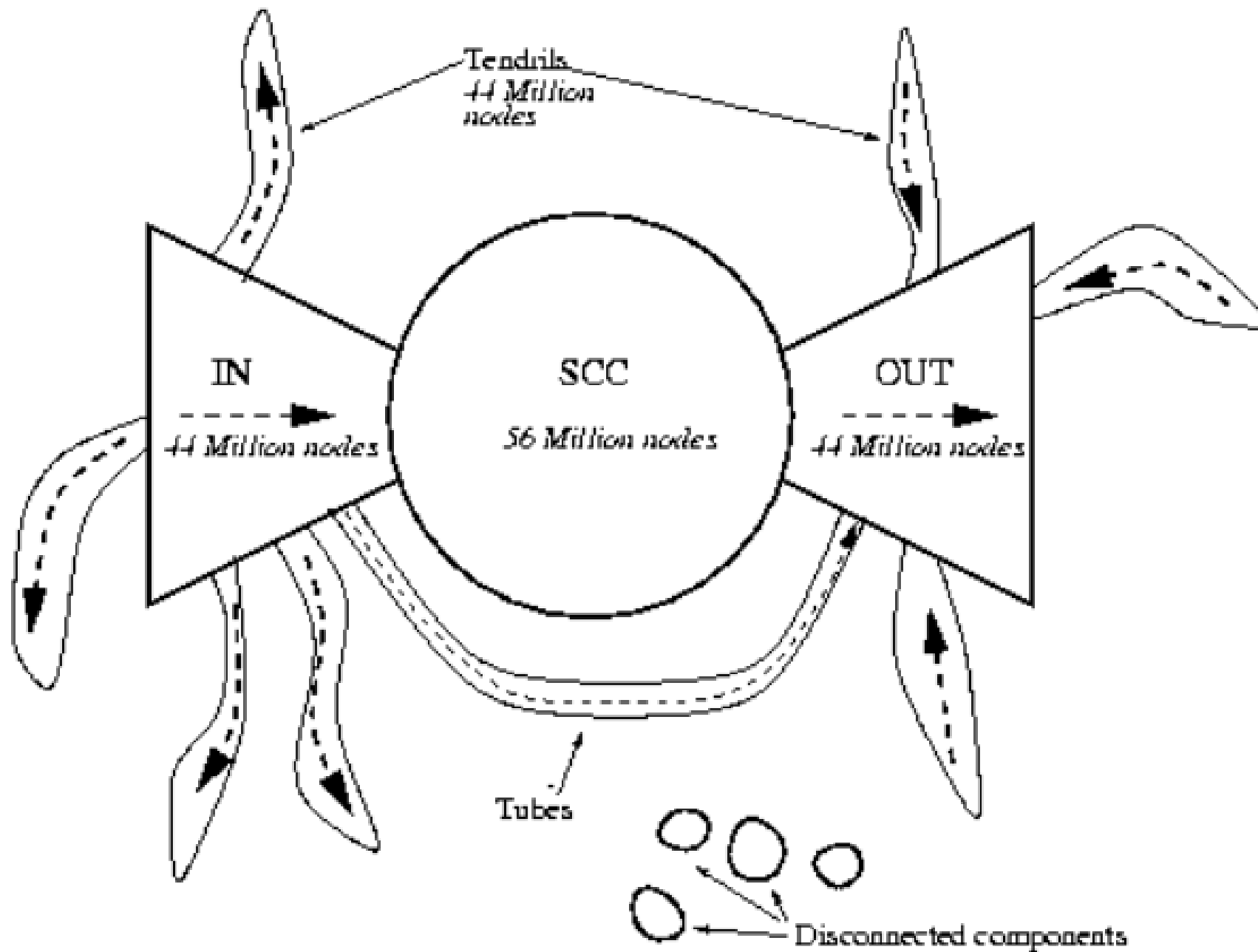
In directed graphs, reachability is **asymmetric**, so connectedness must also be defined asymmetrically.







# Bow-Tie Structure



# Wrap-Up

- short paths alone do not guarantee navigability
- a little randomness can shrink distances dramatically
- information networks need directed notions of path, components, and reachability